# Material-Guided Multiview Fusion Network for Hyperspectral Object Tracking

Zhuanfeng Li, Fengchao Xiong, *Member, IEEE*, Jun Zhou, *Senior Member, IEEE*, Jianfeng Lu, *Member, IEEE*, Zhuang Zhao, and Yuntao Qian, *Senior Member, IEEE*

***Abstract*— Hyperspectral videos (HSVs) have more potential in object tracking than color videos, thanks to their material identification ability. Nevertheless, previous works have not fully explored the benefits of the material information, resulting in limited representation ability and tracking accuracy. To address this issue, this article introduces a material-guided multiview fusion network (MMF-Net) for improved tracking. Specifically, we combine false-color information, hyperspectral information, and material information obtained by hyperspectral unmixing to provide a rich multiview representation of the object. Cross-material attention (CMA) is employed to capture the interaction among materials, enabling the network to focus on the most relevant materials for the target. Furthermore, leveraging the discriminative ability of material view, a novel material-guided multiview fusion module is proposed to capture both intraview and cross-view long-range spatial dependencies for effective feature aggregation. Thanks to the enhanced representation ability of each view and the integration of the complementary advantages of all views, our network is more capable of suppressing the tracking drift in various challenging scenes and achieving accurate object localization. Extensive experiments show that our tracker achieves state-of-the-art tracking performance. The source code will be available at https://github.com/hscv/MMF-Net.**

***Index Terms*— Hyperspectral object tracking, hyperspectral unmixing, multihead attention, multiview fusion.**

## I. Introduction

VISUAL object tracking is an essential task in remote sensing, with numerous applications in unmanned driving and video surveillance [1], [2], just to name a few. Its primary objective is to accurately predict the position of a given target object in video sequences. Recent years have witnessed significant progress in object tracking in grayscale or RGB videos captured by ground-based [3], [4], [5], [6], unmanned aerial vehicle (UAV) [7], or satellite [8], [9], [10] platforms. Nevertheless, object tracking still faces multiple challenges, including background clutter and low resolution. These obstacles are often caused by the limitations of video cameras in providing sufficient spatial information and accurate physical properties of the target object. In contrast, hyperspectral videos (HSVs) record the spatial, spectral, and temporal information simultaneously, which enables the identification of underlying object material and enhances the ability to distinguish objects from background distractors [11], [12], [13]. Accordingly, several progresses have been made on hyperspectral tracking ranging from hand-crafted [11], [14], [15] to deep feature-based approaches [16], [17], [18], [19], [20], [21], [22].

The inherent tradeoff among spatial, spectral, and temporal dimensions causes the low spatial resolution of HSVs, persisting as a huge challenge in hyperspectral tracking. In this case, there exist mixed pixels, especially at the image edges, where the spectral reflectance is a composite of multiple constituent materials. Consequently, the captured spectral information falls short of directly conveying the material details within a scene, leading to challenges in material information exploration [23], [24]. Adding to the complexity, it is known that convolutional networks (CNNs) tend to learn the edge-related features in the first few layers [25]. The low spatial resolution coupled with mixed pixels further brings challenges for discriminative feature learning. Some prior efforts employed hyperspectral unmixing [11], [26], [27], [28] to decompose hyperspectral images (HSIs) into constituent materials (endmembers) and their fractional distributions (abundances) to tackle this issue. These tracking methods, while addressing the issue to some extent, treat all materials equally, disregarding the crucial consideration that not all materials are equally relevant to the target; some are more pertinent to the background than to the object being tracked.

Utilizing multiple features for tracking can create a multiview representation of the target, enabling to leverage of the strengths inherent in different features for enhanced performance. Some efforts have been made in hyperspectral tracking by considering material information as one of the views. For example, the MHT [11] and SiamF [29] trackers incorporated material information using feature concatenation and an additional classifier, respectively. However, these simple interactions among views still fall short of effectively harnessing

Manuscript received 2 June 2023; revised 25 October 2023 and 3 January 2024; accepted 8 February 2024. Date of publication 16 February 2024; date of current version 23 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62002169, Grant 62371237, and Grant 62071421; in part by the "111" Program under Grant B13022; and in part by the Fundamental Research Funds for the Central Universities under Grant 30923010213. *(Corresponding authors: Fengchao Xiong; Jianfeng Lu.)*

Zhuanfeng Li, Fengchao Xiong, and Jianfeng Lu are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: fcxiong@njust.edu.cn; lujf@njust.edu.cn).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia.

Zhuang Zhao is with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

Yuntao Qian is with the College of Computer Science, Zhejiang University, Hangzhou 310027, China.

Digital Object Identifier 10.1109/TGRS.2024.3366536

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

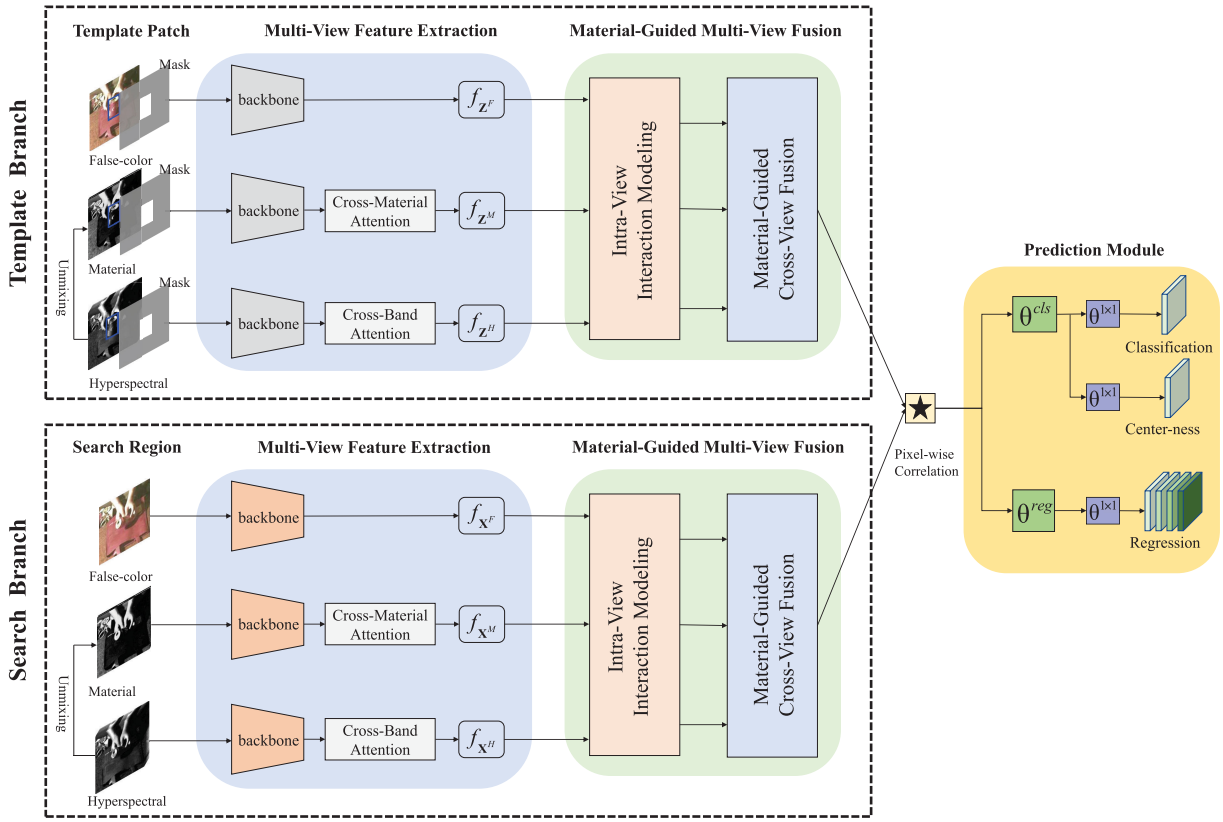



Fig. 1. Framework of our proposed hyperspectral tracker includes three components: a multiview feature extraction module, a material-guided multiview fusion module, and a prediction module.

the complementary information available from different views.

Against these issues, this article introduces a material-guided multiview fusion network (MMF-Net) for hyperspectral object tracking, as illustrated in Fig. 1. The MMF-Net tracker generates a multiview description of the target by augmenting false-color and hyperspectral information with additional material information. All views are processed through a multiview feature extraction module to obtain a deep multiview representation. Given that only certain materials contain strong semantic information relevant to the target, we introduce a cross-material attention (CMA) block to capture interactions between different materials, focusing on more critical materials related to the targeted object. Additionally, for global interaction across bands, we introduce a cross-band attention (CBA) block to enhance the representation of hyperspectral information. Leveraging the distinctive advantages of the material view, we introduce a material-guided multiview fusion module, which employs multihead self-attentions (MHSAs) to model long-range spatial interactions within each view and multihead cross-attentions to capture long-range spatial interactions across views. Consequently, this module effectively encodes specified intraview and cross-view contextual information for more accurate localization. Extensive experiments demonstrate that our MMF-Net tracker significantly outperforms alternative trackers. The ablation study further confirms the effectiveness of the introduced components.

The contributions of this article are summarized as follows.

1) We introduce a CMA module to adaptively focus on critical materials for discriminative tracking. To the best of our knowledge, this is the first time that the varied contributions of different materials have been considered for hyperspectral tracking.
2) We design a novel material-guided multiview fusion module to enhance the feature representation of each view and leverage the multiview complement for a robust joint representation, facilitating more accurate positioning and classification.
3) Extensive experiments show that our tracker achieves the state-of-the-art performance on three hyperspectral benchmarks by an area under the curve (AUC) of **0.691** on the HOTC dataset [11], **0.524** on the HOT2022 dataset [30], and **0.693** on the IMEC25 dataset [31].

The remainder of this article is organized as follows. Section II introduces the related works. Section III gives the details of the proposed tracker. Section IV compares the proposed MMF-Net tracker with alternative trackers. The conclusions are drawn in Section V.

## II. Related Work

### A. Hyperspectral Object Tracking

Hyperspectral trackers can be roughly divided into hand-crafted feature-based and deep-learning-based trackers.

*1) Hand-Crafted Feature-Based Hyperspectral Trackers:* Early hyperspectral object tracking methods [11], [14], [32] rely on hand-crafted features to describe objects. For example, Qian et al. [32] used fixed 3-D cubes as convolution kernels to extract features, while Xiong et al. [11] utilized spectral–spatial histograms of multidimensional gradients

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

(SSHMG) and fractional abundances to extract local–global features. Thanks to the local–global representation, Xiong's method outperforms Qian's by a noticeable margin. Subsequently, Hou et al. [14] replaced SSHMG with the stacked histogram of the oriented gradient feature and used low-rank tensor decomposition to enhance the spectral structure features, boosting the tracking performance. With the assumption that different objects may favor different hyperspectral bands, Zhang et al. [15] proposed an effective band selection strategy based on local image contrast, band entropy, and foreground–background differences to choose the optimal three bands for feature representation, improving both tracking accuracy and speed. However, due to the limited representation ability, hand-crafted features are not robust enough to handle significant appearance changes in complex scenes.

*2) Deep-Learning-Based Hyperspectral Trackers:* Thanks to the powerful representation ability, hyperspectral object tracking has transferred to deep-learning-based methods [19], [20], [21], [33]. Due to limited HSVs, existing works [16], [17], [18], [22], [34] mainly rely on transferring backbone networks trained on color videos for hyperspectral object representation. For instance, HSIs can be converted into a three-channel image and then fed into VGGNet [35] for deep feature extraction [34]. This process, however, results in the loss of valuable spectral information, limiting the tracking performance. To address this challenge, BAE-Net [16] and SiamBAG [22] divide the HSIs into multiple three-channel band groups according to the band importance, which is then used for ensemble tracking. However, they ignore the interactions between different band groups. Gao et al. [36] further improved the tracking performance by employing a bi-directional feature fusion module to integrate features extracted from different band groups and a cross-band group attention module to learn the cross-band interaction. However, these trackers fail to utilize material information, suppressing discriminative ability. In contrast, SiamF [29] incorporated the material information with an additional classifier into hyperspectral information by summarizing the classification maps of different views.

However, SiamF treats all the materials equally during tracking, without considering that some materials may be more closely related to the background rather than the target. Additionally, SiamF does not fully exploit the complementary relationship between material information and other types of information, which compromises the quality of the representation. This article introduces an MMF-Net to address these issues.

### B. Multiview Representation Learning

Multiview representation can learn more comprehensive representations of objects by enjoying the strengths from different views [37], [38]. Multiview representation learning has gained significant attention and can be roughly categorized into two types: multiview representation alignment and multiview representation fusion [37]. Multiview representation alignment learns the relationship between different views using feature alignment methods, such as similarity-based alignment [39], [40] and correlation-based alignment [41], [42]. For example, Song et al. [39] introduced a rank loss based on cosine similarity to align the text and video for cross-modal retrieval. Gao et al. [41] used a deep convolutional encoder network to extract features separately from images and text and then adopted canonical correlation analysis [43] to maximize the correlation between the two features.

Multiview representation fusion aims to integrate information from multiple views to obtain a powerful representation. Instead of the simple feature concatenation, convolutions were employed for a joint representation such as autoencoder-based fusion [44], CNN-based fusion [45], [46], [47], and so on. For instance, Zhang et al. [44] used inner autoencoder networks to encode view-specific information, and outer autoencoder networks to combine all view information for representation. By enforcing the network to focus on the specified locations or channels, the convolutional-based spatial attention [45], [46] and channel attention [48], [49] have been used for multiview representation fusion. For example, Li et al. [45] introduced a 3-D-cross attention block consisting of spatial attention and channel attention to learn a complementary representation of spatial information from HSIs and elevation information from LiDAR. Fazlali et al. [46] introduced CNN-based attention for fusing the Bird's-eye-view and range-view generated from LiDAR data, thus improving the subsequent 3-D object detection performance. However, these methods cannot model the cross-view long-range dependencies. This deficiency may limit the robustness of features in some scenes, such as large-scale variations, where global contextual information is always beneficial to accurate localization [50]. In contrast, multihead attention mechanism [51] has attracted extensive interest in multiview representation fusion, thanks to its nonlocal dependencies exploitation ability and has been successfully applied in many tasks such as joint classification of HSIs and LiDAR data [52], visual grounding [53], visual question answering [54], and more. Zhan et al. [53] introduced the multihead cross-attention to learn the complementary representation by modeling the long-range interrelationship between remote-sensing image and text and achieved remarkable visual grounding performance. Therefore, in this article, we focus on the multihead attention schemes to learn multiview complementary information for robust hyperspectral tracking.

## III. Proposed Method

In this section, we present the details of the proposed material-guided multiview fusion tracker, including the overall architecture, the multiview feature extraction module, the material-guided multiview fusion module, and the prediction module. Moreover, the training and tracking details are provided.

### A. Overall Architecture

As shown in Fig. 1, our tracker has the Siamese architecture, including two branches, that is, the template branch and the search branch. Following [55], our method has unshared parameters in the template branch and search branch. In addition to the false-color and hyperspectral information,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

we also introduce the material information as an auxiliary, yielding a comprehensive multiview description of the object. GoogleNet [56] is used to extract deep multiview features, with shared parameters for all views, to ensure that the extracted features are in the same embedding space and reduce network parameters. The extracted features are then aggregated using the material-guided multiview fusion module, where the material view serves as guidance for fusion. Finally, the prediction module performs pixel-wise correlation on the fused multiview representation of the template branch and search branch to predict the object's position. We provide detailed descriptions of the design principles behind each module in Sections III-B–III-D.

### B. Multiview Feature Extraction Module

The multiview feature extraction module generates deep features to encode each view, with the backbone network, CMA, and CBA.

*1) False-Color View:* The false-color view captures the spatial information of the target. Suppose $\mathbf{X}$ is a hyperspectral frame with $W \times H$ pixels and B bands. We generate the trichromatic image $\mathbf{X}^F$ by converting $\mathbf{X}$ into an image with approximate real colors, following [11]. We then extract deep features $f_{\mathbf{X}^F} \in \mathbb{R}^{d\times h\times w}$ from $\mathbf{X}^F$, where $h$, $w$, and $d$ denote the height, width, and number of channels, respectively.

*2) Hyperspectral View:* The hyperspectral view enables the extraction of valuable spatial–spectral information from the target. However, the lack of data makes it highly challenging to train a neural network directly for HSIs. Therefore, we adopt CNN-based architectures trained on the color images while simultaneously considering the spectral properties of HSIs for representation. Since color images and HSIs are incompatible in terms of the number of channels, we divide the HSIs into $K = B - 2$ groups of three-channel images. We create each group $\mathbf{X}_k$ by combining the adjacent three bands with a stride of one and then feed each group into the backbone GoogleNet to extract corresponding deep features $f_{\mathbf{X}_k^H} \in \mathbb{R}^{d\times h\times w}$. This approach enables us to obtain a local spatial–spectral deep representation of the hyperspectral target effectively.

Extracting deep features independent of individual band groups only encodes local spatial–spectral information but ignores the correlation information between all bands. To address this issue, we introduce a CBA block to enable the extracted features to interact across all bands. The CBA block takes the concatenated deep feature maps of all band groups as input and performs global average pooling (GAP) to generate a vector that captures the global contextual information. This vector is then processed by two fully connected layers followed by the softmax function to learn the nonlinear interaction among different band groups and feature maps. The resulting attention map represents the contribution of each band group for subsequent fusion between all band groups. Mathematically, the attention map generation process is as follows:

$$\mathbf{A}^H = \text{softmax}(\text{FC}_2(\sigma(\text{FC}_1(\text{GAP}(\text{concat}(f_{\mathbf{X}_1^H}, \cdots, f_{\mathbf{X}_K^H})))))). \tag{1}$$

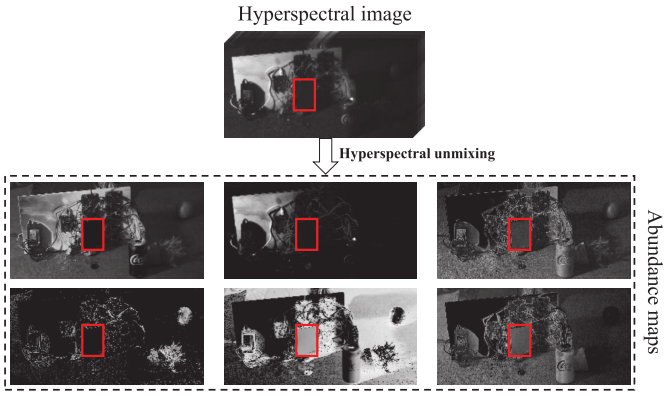



Fig. 2. Visualization of hyperspectral unmixing in a hyperspectral scene.

Here, concat$(\cdot)$ denotes the feature concatenation operation, and $\sigma$ represents the ReLU activation function. $\text{FC}_1$ and $\text{FC}_2$ are two fully connected layers. $\text{FC}_1$ reduces the feature dimensionality from $Kd$ to $(Kd/\tau)$, with $\tau$ being the scale factor, and $\text{FC}_2$ maps the feature dimension back to $K$. Finally, all band groups are fused in a weighted sum manner with the attention map. Mathematically, the calculation procedure is as follows:

$$f_{\mathbf{X}^H} = \sum_{k=1}^{K} \mathbf{A}_k^H f_{\mathbf{X}_k^H}. \tag{2}$$

*3) Material View:* As aforementioned, HSVs usually have a low spatial resolution. The material view provides the distribution of materials within a scene to enhance discrimination, especially in low-resolution and background-cluttered scenarios. The material view is constructed by decomposing each hyperspectral frame into endmembers and abundances, which represent the constituent materials and their fractional distribution, respectively. As shown in Fig. 2, the abundances can provide clearer boundaries with stronger discrimination than the original hyperspectral frame.

Following [11], we assume that the endmembers are fixed for each video and select the endmembers with the collaborative sparse unmixing by variable splitting and augmented Lagrangian (CLSUnSAL) method [57] using the spectral library in the first frame. After that, the abundances of each frame are obtained using the simplex-projection unmixing (SPU) [58] method. To enhance the representation ability, we also use GoogleNet to extract deep features from all the abundance maps.

Moreover, Fig. 2 demonstrates that not all abundance maps are closely associated with the target. Therefore, it is desirable to focus more on the materials that are closely relevant to the target. To achieve this, we propose a CMA block, which enables cross-material interaction and selective fusion of the most important materials for tracking. The process involves computing an attention map $\mathbf{A}^M$ for all abundances and then combining the features of all abundances using the attention maps as weights

$$\begin{aligned}\mathbf{A}^M &= \text{softmax}(\text{FC}_4(\sigma(\text{FC}_3(\text{GAP}(\text{concat}(f_{\mathbf{X}_1^M}, \cdots, f_{\mathbf{X}_P^M})))))) \\ f_{\mathbf{X}^M} &= \textstyle\sum_{p=1}^{P} \mathbf{A}_p^M f_{\mathbf{X}_p^M}\end{aligned} \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

where $f_{\mathbf{X}_p^M} \in \mathbb{R}^{d\times h\times w}$ represents the feature of the $p$th abundance map. $P$ is the total number of abundance maps. $FC_3$ and $FC_4$ are two fully connected layers. $FC_3$ maps the feature dimension to $(Pd/\tau)$ and $FC_4$ maps it to $P$. In this way, we effectively allocate the material information according to its importance.

Up to now, we have obtained the multiview representation of the target, including the false-color view for spatial information $f_{\mathbf{X}^F}$, the hyperspectral view for spatial–spectral information $f_{\mathbf{X}^H}$, and the material view for detailed material information $f_{\mathbf{X}^M}$. However, currently, all views are depicted independently and cannot fully benefit from the complementary information across views. Therefore, we introduce the material-guided multiview fusion module to enable all views to collaborate and enjoy complementary information from each other. This manner can yield a more powerful representation for improved tracking robustness.

### C. Material-Guided Multiview Fusion Module

The material-guided multiview fusion module performs intraview interaction modeling (IVIM) and material-guided cross-view fusion (MGCVF) blocks to enable multiview complementarity.

*1) Intraview Interaction Modeling:* Before multiview fusion, we first enhance the discrimination of each view by an IVIM block. Global contexts are beneficial for understanding scenes. As shown in Fig. 3, we achieve the long-range spatial dependency modeling with the MHSA, which runs through a self-attention mechanism several times in parallel, with each one building the pair-wise dependencies across the tokens.

Given the queries $\mathbf{Q} \in \mathbb{R}^{N\times d}$, keys $\mathbf{K} \in \mathbb{R}^{N\times d}$, and values $\mathbf{V} \in \mathbb{R}^{N\times d}$, the scale dot-production-based attention can be computed by

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}. \tag{4}$$

By projecting $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ for $h$ times with different parameters $\mathbf{W}_h^Q \in \mathbb{R}^{d\times d_k}$, $\mathbf{W}_h^K \in \mathbb{R}^{d\times d_k}$, $\mathbf{W}_h^V \in \mathbb{R}^{d\times d_v}$, the multihead attention is computed as follows:

$$\begin{aligned}\text{MultiHead}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \text{concat}(\text{Head}_1,\cdots,\text{Head}_h)\mathbf{W}^O\\ \text{Head}_i &= \text{Attention}\left(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V\right)\end{aligned} \tag{5}$$

where $\mathbf{W}^O \in \mathbb{R}^{hd_v\times d}$ represents the learnable matrices for further projection. Based on the multihead attention, we can build the MHSA within the given flattened feature map $\mathbf{F} \in \mathbb{R}^{N\times d}$ via

$$\text{MHSA}(\mathbf{F}) = \mathbf{F} + \text{MultiHead}(\mathbf{F}+\mathbf{P},\mathbf{F}+\mathbf{P},\mathbf{F}) \tag{6}$$

where $\mathbf{P} \in \mathbb{R}^{N\times d}$ is the positional encoding to account for the position loss of the flattened feature maps. We use the sine function [59] to encode the position.

Accordingly, the intraview global spatial interaction can be modeled as follows:

$$\begin{aligned}f_{\mathbf{X}^F} &= \text{MHSA}(\text{flatten}(f_{\mathbf{X}^F}))\\ f_{\mathbf{X}^H} &= \text{MHSA}(\text{flatten}(f_{\mathbf{X}^H}))\\ f_{\mathbf{X}^M} &= \text{MHSA}(\text{flatten}(f_{\mathbf{X}^M}))\end{aligned} \tag{7}$$

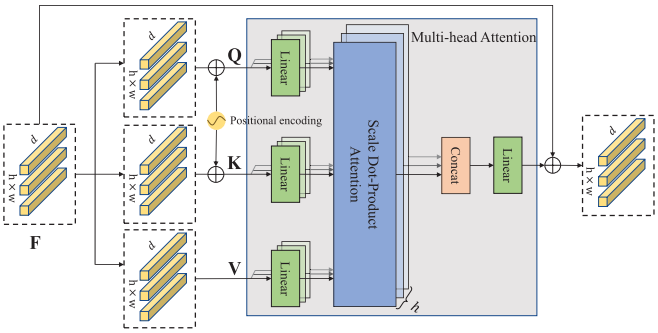



Fig. 3. IVIM based on MHSA.

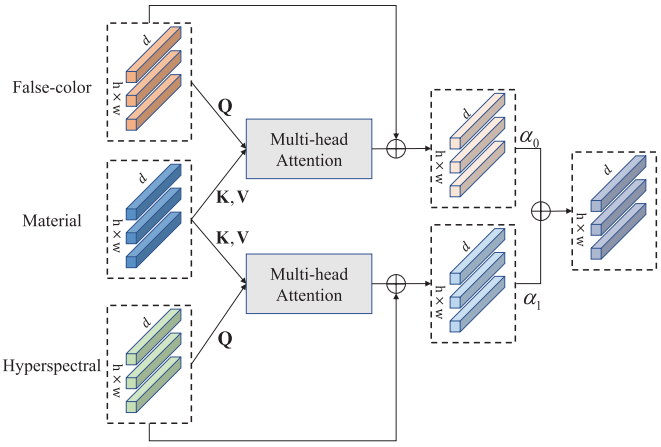



Fig. 4. MGCVF block based on multihead cross-attention.

where flatten$(\cdot)$ transforms the input feature map $\mathbb{R}^{d\times h\times w}$ in spatial dimension to feature sequences $\mathbb{R}^{hw\times d}$. Consequently, the intraview representation is dramatically strengthened.

*2) Material-Guided Cross-View Fusion:* To further promote the global spatial interaction across views, we propose an MGCVF block. This block combines the material view with the false-color and hyperspectral views, using the material view as guidance. The main reason is that the material view provides the unmixing results of given low-resolution HSIs, which contain more discriminative semantics with clearer boundaries, as can be seen from Fig. 2.

As shown in Fig. 4, we use the material view to provide the keys and values and other views as the queries. As an example, let us consider the fusion of the material view and hyperspectral view, which is formulated as follows:

$$\text{MHCA}(f_{\mathbf{X}^H}, f_{\mathbf{X}^M}) = f_{\mathbf{X}^H} + \text{MultiHead}(f_{\mathbf{X}^H}+\mathbf{P}_1,\ f_{\mathbf{X}^M}+\mathbf{P}_2,\ f_{\mathbf{X}^M}) \tag{8}$$

where $\mathbf{P}_1 \in \mathbb{R}^{hw\times d}$ and $\mathbf{P}_2 \in \mathbb{R}^{hw\times d}$ represent the spatial positional encoding of hyperspectral view and material view, respectively. Similarly, the fusion of material view and false-color view can also be conducted with (8).

With the above steps, the global spatial interactions across two different views are learned but interactions among three views are ignored. Therefore, we further combine the fusion results weighted by $\alpha$, that is,

$$\mathbf{F} = \alpha_0\text{MHCA}(f_{\mathbf{X}^H}, f_{\mathbf{X}^M}) + \alpha_1\text{MHCA}(f_{\mathbf{X}^F}, f_{\mathbf{X}^M}). \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

Here, $\alpha$ is formulated by

$$\alpha_i = \frac{e^{w_i}}{e^{w_0} + e^{w_1}} \quad \text{for } i \in \{0, 1\} \tag{10}$$

where $\mathbf{w} = [w_0, w_1]$ is a learnable vector. In this way, we can balance the contribution of fused views for enhanced representation. In our experiment, $[w_0, w_1]$ are both initially set to 1.

### D. Prediction Module

In accordance with [55], the prediction module calculates pixel-level similarities $\mathbf{M} \in \mathbb{R}^{hw \times hw}$ to model the correlation between the feature $\mathbf{F}_Z \in \mathbb{R}^{hw \times d}$ from template patch and the feature $\mathbf{F}_X \in \mathbb{R}^{hw \times d}$ from the search region. The matrix $\mathbf{M}$ is obtained by computing the softmax of the dot product between $\mathbf{F}_Z$ and $\mathbf{F}_X$, divided by the square root of $d$ to prevent numerical errors

$$\mathbf{M} = \text{softmax}\left(\frac{\mathbf{F}_X \mathbf{F}_Z^T}{\sqrt{d}}\right). \tag{11}$$

Based on the pixel-level similarity matrix, we retrieve the target-related information from the template via

$$\mathbf{F}'_X = \mathbf{M}\mathbf{F}_Z. \tag{12}$$

Finally, we concatenate $\mathbf{F}'_X$ with $\mathbf{F}_X$ along the channel dimension to obtain $\mathbf{Y}$ for location prediction.

Afterward, we utilize an anchor-free prediction head that consists of a classification subnetwork and a regression subnetwork to locate the object. The classification subnetwork performs binary classification on each pixel of $\mathbf{Y}$ to create a predicted classification map indicating the likelihood of the target object. The classification subnetwork applies a seven-layer convolutional network, with each layer comprising 256 convolutions with $3 \times 3$ kernel, followed by one convolution with a $1 \times 1$ kernel to map $\mathbf{Y}$ into the classification response map. To overcome unreliable prediction results near the object boundary, another $1 \times 1$ convolution generates a center-ness map after the CNN of the classification subnetwork. When tracking, we perform element-wise multiplication between the center-ness and classification map to discourage the classification score of the positions far from the object center. The regression subnetwork estimates the bounding box on each location. Similar to the classification subnetwork, the regression subnetwork employs another CNN with the same configuration as the classification branch, followed by four convolutions with a $1 \times 1$ kernel to generate the regression map.

### E. Training and Tracking

*1) Offline Training:* To obtain image patches for training, we crop a square patch that is four times the size of the object's bounding box from both the hyperspectral and false-color frames. These patches are then resized to $289 \times 289$ following [55] to generate the template patch and search region. The large cropping size can capture rich contextual information, but the resulting template patch may contain too much background information. Thus, it is necessary to introduce an object mask to tell the foreground and background. Accordingly, we use the ground-truth (GT) bounding box of the template patch to obtain an object mask telling the foreground and background regions. Both the object mask and the template patch are fed into the feature extraction module. Specifically, the template patch passes through the first convolutional layer of GoogleNet to obtain the shallow feature and the mask passes through another convolutional network with 32 convolutions and $3 \times 3$ kernel. Subsequently, the features of template patch and object mask are element-wise added, whose results are passed through the remaining layers of the GoogleNet.

To determine which locations in the classification map are positive samples, we compare the GT bounding box with each location $(p_i, p_j)$ in the map. We remap the location $(p_i, p_j)$ in the classification map back to the search region. The remapping process can be expressed as $(\hat{p}_i, \hat{p}_j) = (\lfloor (s/2) \rfloor + p_i s, \lfloor (s/2) \rfloor + p_j s)$, where $s = 8$ represents the total stride of backbone. If the corresponding position $(\hat{p}_i, \hat{p}_j)$ in the search region falls inside the GT bounding box, the location $(p_i, p_j)$ in classification map is considered a positive sample; otherwise, it is a negative sample. For regressing the target bounding box, we use only positive samples from the locations in the classification map, that is,

$$\begin{aligned} l &= \hat{p}_i - x_l, t = \hat{p}_j - y_l \\ r &= x_r - \hat{p}_i, b = y_r - \hat{p}_j \end{aligned} \tag{13}$$

where $(x_l, y_l)$ and $(x_r, y_r)$ represent the left-top and right-bottom corners of GT, respectively. $(l, t, r, b)$ represents the distances from the location $(\hat{p}_i, \hat{p}_j)$ to four sides of the GT.

The loss function of MMF-Net is a multitask loss including a focal loss for classification [60], a binary cross-entropy loss for center-ness quality assessment, and an IOU loss [61] for regression

$$\begin{aligned} \mathcal{L} = &\frac{\lambda_1}{N_{\text{pos}}} \sum_{i,j} \mathcal{L}_{\text{focal}}\left(p^*_{i,j}, p_{i,j}\right) \\ &+ \frac{\lambda_2}{N_{\text{pos}}} \sum_{i,j} \mathbf{1}_{\{p_{i,j}>0\}} \mathcal{L}_{\text{ce}}\left(q^*_{i,j}, q_{i,j}\right) \\ &+ \frac{\lambda_3}{N_{\text{pos}}} \sum_{i,j} \mathbf{1}_{\{p_{i,j}>0\}} \mathcal{L}_{\text{iou}}\left(o^*_{i,j}, o_{i,j}\right) \end{aligned} \tag{14}$$

where $\lambda_1 = 1, \lambda_2 = 1$, and $\lambda_3 = 3$ balance three losses. $\mathbf{1}_{\{p_{i,j}>0\}}$ is 1 if the condition $p_{i,j} > 0$ holds and 0 if not. The $N_{\text{pos}}$ represents the number of positive samples. $p_{i,j}$ refers to the label of classification with the value of 1 if the location $(p_i, p_j)$ is in a positive sample and 0 in a negative sample. $q_{i,j} = ((\min(l, r)/\max(l, r)) \times (\min(t, b)/\max(t, b)))^{1/2}$ represents the label of center-ness [62]. $o_{i,j}$ represents the label of regression, that is, the coordinate in the left-top and right-bottom corners of the GT bounding box. $p^*_{i,j}$, $q^*_{i,j}$, and $o^*_{i,j}$, respectively, represent the predicted values of classification, center-ness, and regression from the prediction module. In (14), the detailed loss functions can be expressed as follows:

$$\begin{aligned} &\mathcal{L}_{\text{focal}}\left(p^*_{i,j}, p_{i,j}\right) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \\ &\mathcal{L}_{\text{ce}}\left(q^*_{i,j}, q_{i,j}\right) = -q_{i,j} \log\left(q^*_{i,j}\right) - \left(1 - q_{i,j}\right) \log\left(1 - q^*_{i,j}\right) \\ &\mathcal{L}_{\text{iou}}\left(o^*_{i,j}, o_{i,j}\right) = -\ln\left(o^*_{i,j} \cap o_{i,j} / o^*_{i,j} \cup o_{i,j}\right). \end{aligned} \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

In the focal loss, if $p_{i,j}$ is 1, then $\alpha_t$ is $\alpha$ and $p_t$ is $p^*_{i,j}$; if $p_{i,j}$ is 0, then $\alpha_t$ is $1-\alpha$ and $p_t$ is $1-p^*_{i,j}$. $\alpha$ addresses the class imbalance, and $\gamma$ represents the focusing parameter to focus more on hard and misclassified examples. In the experiment, $\alpha$ is set as 0.75, while $\gamma$ is set as 2. In IOU loss, $\cap(\cdot)$ calculates the intersection area between the predicted bounding box and the GT bounding box, and $\cup(\cdot)$ calculates the union area between them.

*2) Online Tracking:* We start the tracking by cropping the first frame to obtain a template patch and generate an object mask based on the initial state of the object. The object mask and the template patch are fed into the multiview feature extraction module to generate the multiview features, which are then combined by a material-guided multiview fusion module to obtain the feature $\mathbf{F}_Z$. In the subsequent frames, the search region is cropped to extract the feature $\mathbf{F}_X$ according to the predicted state in the previous frame. $\mathbf{F}_Z$ and $\mathbf{F}_X$ then pass through the prediction module to output the classification map, center-ness map, and regression map. To encourage the classification scores near the target center and suppress the scores away from the center, the center-ness map is multiplied with the classification map to produce a new classification map with better localization accuracy. Finally, the object location is predicted by the following equation:

$$\begin{aligned} x^*_l &= \hat{p}^*_i - l^*_{ij}, \; y^*_l = \hat{p}^*_j - t^*_{ij} \\ x^*_r &= \hat{p}^*_i + r^*_{ij}, \; y^*_r = \hat{p}^*_j + b^*_{ij} \end{aligned} \tag{16}$$

where $(x^*_l, y^*_l)$ and $(x^*_r, y^*_r)$ represent the coordinates of the top-left and the bottom-right corners of the predicted bounding box. The location $(\hat{p}^*_i, \hat{p}^*_j)$ is obtained by remapping the location $(p^*_i, p^*_j)$ in the new classification map with the highest classification score to the input search region. The $l^*, t^*, r^*, and b^*$ denote predicted distance in the regression maps, respectively.

## IV. Experiments

In this section, we first compare our tracker MMF-Net with state-of-the-art trackers to show its superior tracking ability. Moreover, we conduct an ablation study to analyze the effectiveness and impact of different components on performance.

### A. Experimental Setup

Before the performance comparison, we first present the network implementation details, evaluation indices, dataset description, and compared methods.

*1) Network Implementation:* Our MMF-Net was implemented in Python and trained on an NVIDIA Tesla V100 GPU. The backbone and prediction head were pretrained on the color videos, following [55]. We kept these parameters fixed and trained only the other parameters in our implementation. Specifically, the network was trained for 20 epochs with a batch size of 16. The momentum and weight decay rate were set to 0.9 and $1\times10^{-4}$, respectively. We adopted a warming-up strategy by increasing the learning rate from $1 \times 10^{-6}$ to $5 \times 10^{-4}$ in the first epoch. Thereafter, we used a cosine annealing learning rate schedule to decrease the learning rate from $5 \times 10^{-4}$ to $1 \times 10^{-6}$ for the remaining 19 epochs. The value of $\tau$ was set to 16 in both the CBA block and the CMA block. The head number $h$ of multihead attention was set to 8.

*2) Evaluation Indices:* Following [63], we evaluate the tracking performance using two widely adopted protocols: precision plot and success plot. The success plot calculates the fraction of frames where the overlap between the GT and predicted bounding box exceeds a predefined threshold. The precision plot records the percentage of frames with a center position error below the specified threshold. Our evaluation employs a one-pass evaluation (OPE) to assess tracking results. To rank the compared trackers, we select the AUC of the success plots and measure distance precision (DP) using a threshold of 20 pixels.

*3) Hyperspectral Datasets:* In the experiment, we use three hyperspectral datasets to evaluate our method. The detailed description is as follows.

*HOTC Dataset:* HOTC [11] is the public hyperspectral dataset in the first Hyperspectral Object Tracking Challenge. HOTC contains 40 video sequences for training and 35 video sequences for testing. The training part has a total of 13 000 annotated frames, while the testing part has a total of 16 500 annotated frames. The frame rate of HOTC is 25 frames/s. Each video includes the HSV, color video, and false-color video converted from HSVs under the same scenes. Each frame of the HSV contains 16 bands in the wavelength from 470 to 620 nm. The dataset is labeled with 11 different attributes corresponding to different scenarios. The attributes include illumination variation (IV), in-plane rotation (IPR), motion blur (MB), background clutter (BC), occlusion (OCC), scale variation (SV), out-of-view (OV), out-of-plane rotation (OPR), deformation (DEF), low resolution (LR), and fast motion (FM).

*HOT2022 Dataset:* HOT2022 [30] is a public dataset for evaluating hyperspectral object trackers in the second Hyperspectral Object Tracking Challenge. The dataset consists of 45 video sequences with a total of 28 000 frames. The dataset enriches the testing part of the HOTC with diverse categories of objects and complex background surroundings. Besides 11 attributes in the HOTC dataset, an additional camera motion (CM) attribute is included. All other settings are the same as those in the HOTC dataset.

*IMEC25 Dataset:* The IMEC25 [31] dataset consists of 55 HSVs for training and 80 HSVs for testing. All training videos contain approximately 14 000 annotated frames, and the testing videos contain approximately 14 060 annotated frames. Each frame in the HSVs contains 25 wavelength bands ranging from 680 to 960 nm. The frame rate of the IMEC25 dataset is 10 frames/s. Similar to the HOTC dataset, the IMEC25 dataset is also annotated with 11 different attributes, corresponding to different challenging scenarios.

*4) Compared Methods:* Nine methods are selected to validate the effectiveness of our method, including CNHT [32], DeepHKCF [34], MHT [11], BAE-Net [16], MFI [64], SSDT-Net [19], SST-Net [17], SiamF [29], and SiamBAG [22]. The DeepHKCF, BAE-Net, SST-Net, SSDT-Net, SiamF, and SiamBAG employ deep features, while the others employ hand-crafted features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I

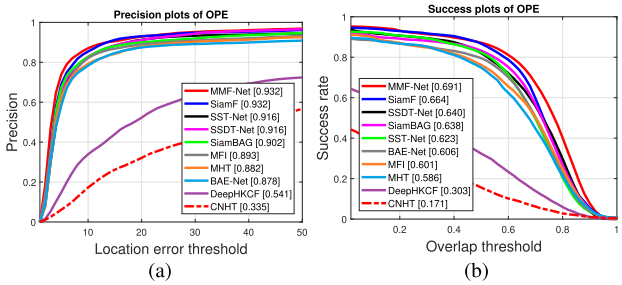
ATTRIBUTE-BASED COMPARISON ON THE HOTC DATASET. THE TOP TWO VALUES ARE HIGHLIGHTED BY **RED** AND **BLUE**

| Attribute | MMF-Net | SiamBAG [22] | SiamF [29] | SSDT-Net [19] | SST-Net [17] | BAE-Net [16] | MFI [64] | MHT [11] | DeepHKCF [34] | CNHT [32] |
|---|---|---|---|---|---|---|---|---|---|---|
| BC | **0.708** | 0.653 | **0.698** | 0.663 | 0.685 | 0.651 | 0.651 | 0.606 | 0.284 | 0.183 |
| DEF | **0.724** | 0.690 | **0.716** | 0.693 | 0.700 | 0.680 | 0.639 | 0.664 | 0.426 | 0.288 |
| FM | **0.704** | 0.611 | **0.703** | 0.603 | 0.561 | 0.608 | 0.600 | 0.542 | 0.264 | 0.176 |
| IV | **0.592** | 0.543 | 0.578 | **0.580** | 0.502 | 0.525 | 0.516 | 0.478 | 0.129 | 0.096 |
| IPR | **0.782** | 0.712 | **0.733** | 0.674 | 0.696 | 0.699 | 0.692 | 0.670 | 0.485 | 0.272 |
| LR | **0.627** | 0.583 | **0.650** | 0.559 | 0.462 | 0.489 | 0.514 | 0.476 | 0.082 | 0.027 |
| MB | **0.764** | 0.648 | **0.710** | 0.571 | 0.536 | 0.593 | 0.570 | 0.560 | 0.363 | 0.094 |
| OCC | **0.650** | 0.601 | 0.615 | **0.630** | 0.594 | 0.555 | 0.546 | 0.565 | 0.250 | 0.118 |
| OPR | **0.769** | 0.697 | **0.719** | 0.704 | 0.699 | 0.701 | 0.680 | 0.644 | 0.428 | 0.259 |
| OV | 0.602 | **0.636** | 0.621 | **0.647** | 0.480 | 0.517 | 0.611 | 0.626 | 0.286 | 0.144 |
| SV | **0.680** | 0.632 | **0.651** | 0.642 | 0.602 | 0.604 | 0.599 | 0.574 | 0.298 | 0.156 |

TABLE II

PERFORMANCE COMPARISON IN TERMS OF COMPUTATIONAL COMPLEXITY. THE TOP TWO VALUES ARE HIGHLIGHTED BY **RED** AND **BLUE**

| Trackers | MMF-Net | SiamBAG [22] | SiamF [29] | SST-Net [17] | BAE-Net [16] | MFI [64] | MHT [11] | DeepHKCF [34] | CNHT [32] |
|---|---|---|---|---|---|---|---|---|---|
| FPS | **6.11** | **10.52** | 2.96 | 0.65 | 0.72 | 0.42 | 2.23 | 0.91 | 2.61 |
| FLOPs(G) | 352.6 | **126.3** | **227.7** | 315.0 | 314.9 | - | - | - | - |
| Devices | GPU | GPU | GPU | GPU | GPU | CPU | CPU | CPU | CPU |




(a) (b)

Fig. 5. Precision and success plots of all compared hyperspectral trackers on the HOTC dataset. (a) Precision plot. (b) Success plot.

### B. Comparison With State-of-the-Art Hyperspectral Trackers

*1) Evaluation on the HOTC Dataset:* We provide a quantitative comparison of all the compared methods in Fig. 5. We observe that the performance of hand-crafted feature-based trackers is generally lower than that of deep feature-based trackers. This is mainly due to the inherent limitations of hand-crafted features for discriminative representation, which restricts their ability to handle various challenging scenarios. The material information can provide more distinguishable visual clues and thereby help SiamF outperform SST-Net, BAE-Net, and SiamBAG to a certain margin with respect to all the indexes.

Notably, our MMF-Net ranks top among all the compared trackers and achieves an AUC of **0.691** and a DP of **0.932**. In particular, the superior performance of MMF-Net is attributed to two aspects. On the one hand, the exploitation of multiview representation with cross-band and CMA enables a comprehensive description of the hyperspectral object. On the other hand, the material-guided multiview fusion module helps depict the long-range intraview and cross-view interactions, allowing for the sharing of complementary information across views. As a result, the generated representation is more robust and can handle various challenging scenarios effectively. Additionally, we present attribute-based evaluations in terms of 11 annotated attributes. As shown in Table I, our MMF-Net achieves the best performance on nine out of 11 attributes, further demonstrating its effectiveness.

In Fig. 6, we present a visual comparison of our MMF-Net tracker with the top seven hyperspectral trackers on six typical sequences from the HOTC dataset. Overall, our MMF-Net tracker achieves more precise predictions. The lack of material information limits the robustness of predictions in complex scenes, resulting in BAE-Net and SST-Net failing to accurately locate the target object in the *car3* and *student* sequences. Thanks to the complement between multiview information, our MMF-Net provides more accurate localization. Moreover, in the *toy1* sequence, SiamF fails to predict the target object accurately due to the lack of multiview interactions. Our method outperforms the compared trackers in these sequences due to the effective intraview and cross-view exploitation of objects.

We additionally recorded the computational complexity of all the methods in terms of floating-point operations per second (FLOPs) and frames per second (FPS). The compared trackers include hand-crafted feature-based trackers and deep feature-based trackers. The hand-crafted feature-based trackers ran on a Windows machine with the configuration of an Intel[1] Core[2] i5-4440 CPU@3.10 GHz and 16 GB RAM, while the deep-learning-based trackers ran on a machine equipped with two Intel[1] Xeon[2] Silver 4114 CPUs@2.20 GHz and an NVIDIA Tesla V100 GPU. SSDT-Net is omitted because of the absence of its codes.

As shown in Table II, among all the trackers, our MMF-Net requires the highest number of FLOPs due to feature extraction from all views. However, in terms of FPS, the MMF-Net is faster than all the trackers except for SiamBAG. In real-world applications, tracking accuracy is often prioritized over speed

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

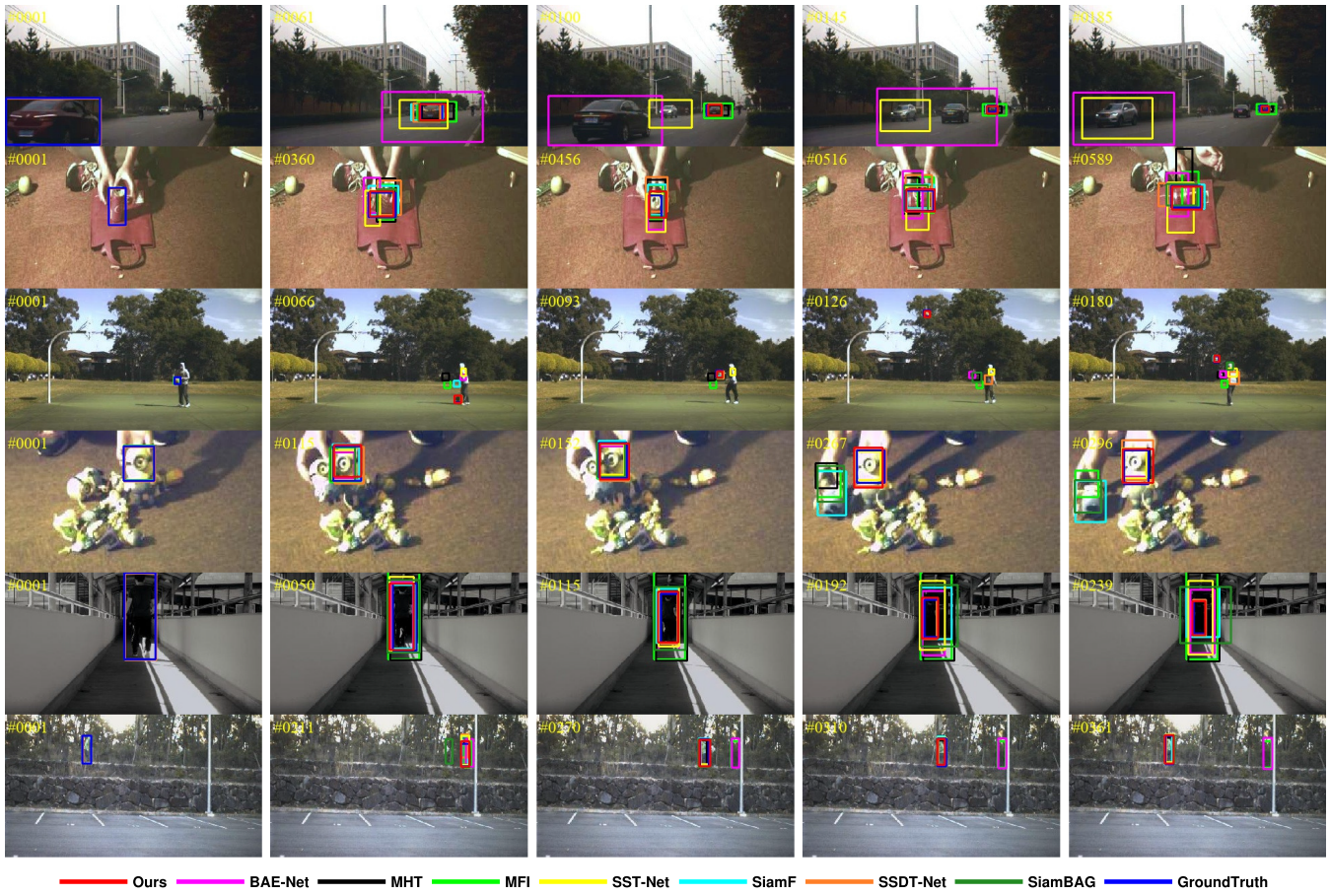
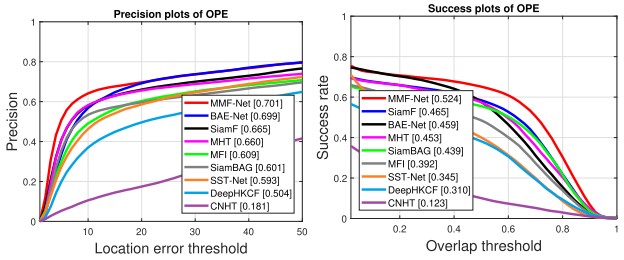



Fig. 6. Qualitative comparison of the MMF-Net tracker on six video sequences (*car3, coke, basketball, toy1, student, and forest2*) on the HOTC dataset. The proposed approach performs favorably against these trackers (for better illustration, we show the bounding boxes on the false-color image converted from HSIs).




(a) (b)

Fig. 7. Precision and success plots of all the compared hyperspectral trackers on the HOT2022 dataset. (a) Precision plot. (b) Success plot.

and computational resources. Our MMF-Net provides a good balance between tracking performance and efficiency.

*2) Evaluation on the HOT2022 Dataset:* In Fig. 7, we provide a performance comparison of all the trackers on the HOT2022 dataset to gain more insights into the effectiveness of our proposed approach. All the trackers use the parameters set on the HOTC dataset for testing to evaluate their generalization ability. We exclude SSDT-Net due to the absence of its code. Our MMF-Net still outperforms the other compared methods by a significant margin, achieving the highest AUC of **0.524** and DP of **0.701**. This superior performance once again demonstrates the robustness of our MMF-Net tracker. Moreover, we also provide the performance in terms of 11 challenging attributes in Table III. The MMF-Net tracker achieves top-two performance in all challenging scenes. Our tracker encodes each view using cross-band and CMAs while leveraging intraview and cross-view long-range spatial interactions to achieve complementary benefits, leading to improved overall performance. Furthermore, we visually compare the tracking results on the six typical sequences in Fig. 8. As shown in the figure, our tracker achieves the optimal performance.

*3) Evaluation on the IMEC25 Dataset:* In this experiment, we compare all the methods except SSDT-Net on the IMEC25 dataset. As the SiamF, SST-Net, BAE-Net, SiamBAG, and MMF-Net methods were originally trained on the HOTC dataset with 16 bands and are not suitable for the IMEC25 dataset with 25 bands, we retrained them following the original settings. Since the IMEC25 dataset does not provide false-color videos, we choose the three bands indexed by [1, 13, 25] from HSIs to generate the false-color image. As shown in Fig. 9, the proposed MMF-Net shows very promising tracking performance and outperforms the second-best tracker SiamF with a gain of **0.035** and **0.019** in terms of AUC and DP scores, respectively. The promising performance is attributed to the effective multiview feature extraction and material-guided multiview fusion, which provide an effective way to leverage the complementarity of multiview information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

TABLE III

ATTRIBUTE-BASED COMPARISON ON THE HOT2022 DATASET. THE TOP TWO VALUES ARE HIGHLIGHTED BY **RED** AND **BLUE**

| Attribute | MMF-Net | SiamBAG [22] | SiamF [29] | SST-Net [17] | BAE-Net [16] | MFI [64] | MHT [11] | DeepHKCF [34] | CNHT [32] |
|---|---|---|---|---|---|---|---|---|---|
| BC | **0.493** | 0.412 | 0.410 | 0.292 | **0.432** | 0.352 | 0.428 | 0.308 | 0.125 |
| DEF | **0.579** | 0.481 | 0.531 | 0.374 | 0.488 | 0.389 | **0.538** | 0.300 | 0.116 |
| FM | **0.305** | 0.249 | 0.297 | 0.271 | **0.330** | 0.212 | 0.241 | 0.190 | 0.049 |
| IPR | **0.479** | 0.438 | 0.412 | 0.330 | **0.497** | 0.439 | 0.448 | 0.372 | 0.133 |
| IV | **0.744** | 0.698 | **0.730** | 0.549 | 0.618 | 0.485 | 0.571 | 0.364 | 0.117 |
| LR | **0.323** | 0.186 | **0.243** | 0.154 | 0.172 | 0.171 | 0.163 | 0.088 | 0.074 |
| MB | **0.319** | 0.227 | **0.269** | 0.155 | 0.257 | 0.182 | 0.221 | 0.116 | 0.066 |
| OCC | **0.460** | 0.361 | 0.401 | 0.306 | 0.392 | 0.335 | **0.404** | 0.238 | 0.098 |
| OPR | **0.557** | 0.507 | **0.530** | 0.337 | 0.512 | 0.466 | 0.526 | 0.377 | 0.171 |
| OV | **0.613** | 0.564 | **0.576** | 0.482 | 0.541 | 0.483 | 0.500 | 0.393 | 0.163 |
| SV | **0.578** | 0.507 | 0.519 | 0.389 | **0.547** | 0.456 | 0.521 | 0.363 | 0.148 |
| CM | **0.563** | 0.490 | **0.510** | 0.372 | 0.486 | 0.429 | 0.508 | 0.369 | 0.136 |

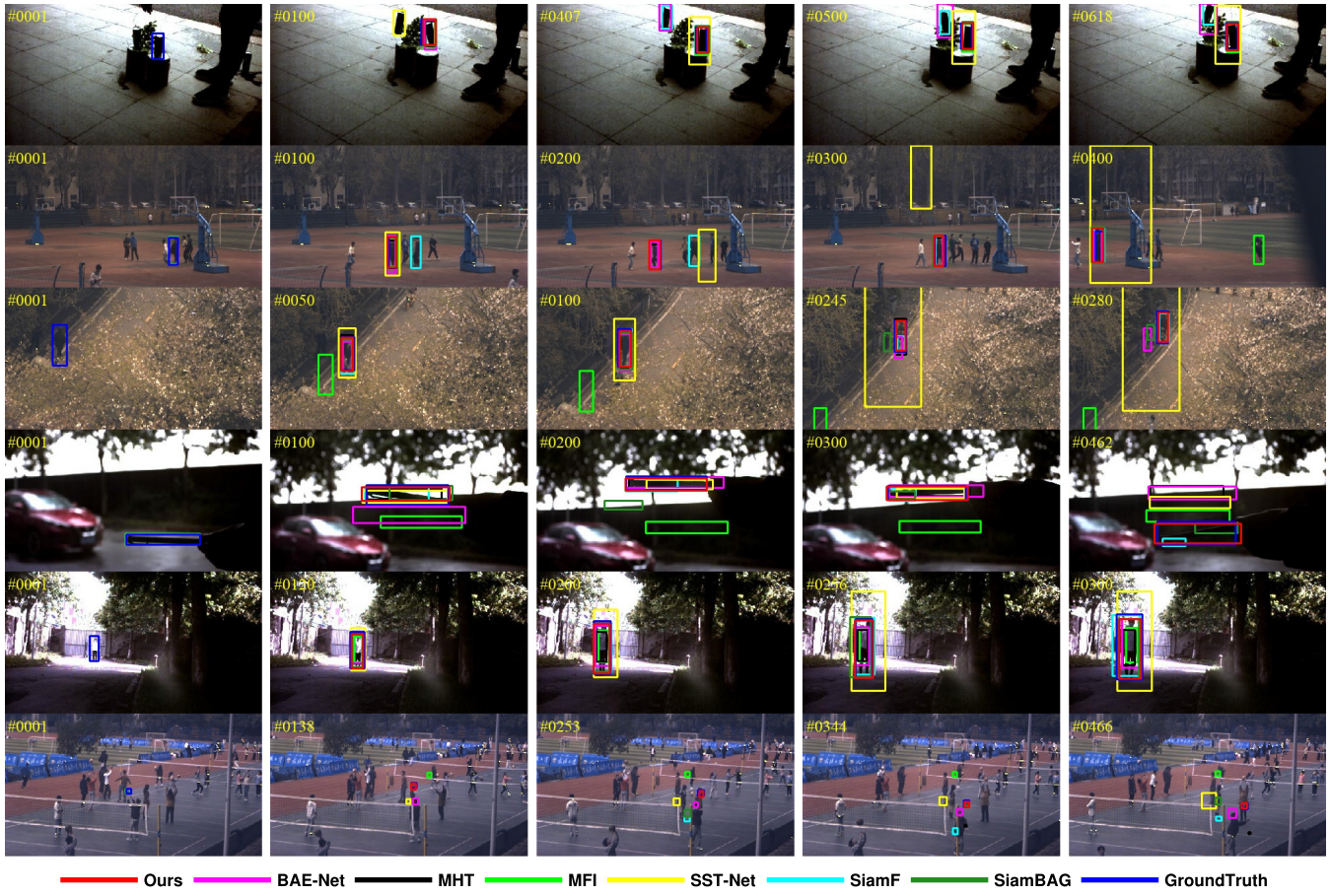


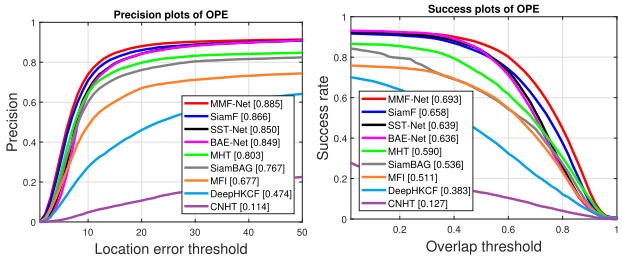

Fig. 8. Qualitative comparison of the MMF-Net tracker on six video sequences (*cup5, L_basketball_person, L_person7, pen2, people3, and S_volleyball1*) on the HOT2022 dataset. The proposed approach performs favorably against these trackers (for better illustration, we show the bounding boxes on the false-color image converted from HSIs).

Furthermore, the attribute-based evaluation is summarized in Table IV. Our MMF-Net tracker achieves the best performance on eight out of the 11 attributes, except for BC, OCC, and IV, where it achieved the second-best performance. In the LR scenario, our MMF-Net tracker performs better than the SST-Net and BAE-Net with a gain of 0.022 and 0.041 in AUC score because our tracker selectively considers material information along with hyperspectral information, which provides more discriminative information. Additionally, in the IPR, MB, SV, and OV scenarios, the SiamF and MMF-Net trackers achieve significant gains in the AUC score by using multiview information. Our method surpasses SiamF in most scenarios due to the effective multiview fusion module that fully exploits the complementary information between different views. In summary, the experiments on the IMEC25 dataset provide clear evidence that our MMF-Net is highly proficient in representing and combining different views, resulting in significantly improved tracking performance. Moreover, we further present the visual results on the six typical sequences from the IMEC25 dataset in Fig. 10 and our method still provides the most robust tracking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ATTRIBUTE-BASED COMPARISON ON THE IEMC25 DATASET. THE TOP TWO VALUES ARE HIGHLIGHTED BY **RED** AND **BLUE**

| Attribute | MMF-Net | SiamBAG [22] | SiamF [29] | SST-Net [17] | BAE-Net [16] | MFI [64] | MHT [11] | DeepHKCF [34] | CNHT [32] |
|---|---|---|---|---|---|---|---|---|---|
| BC | **0.617** | 0.484 | 0.592 | **0.648** | 0.616 | 0.517 | 0.586 | 0.340 | 0.101 |
| DEF | **0.715** | 0.567 | 0.678 | 0.677 | **0.689** | 0.507 | 0.614 | 0.326 | 0.067 |
| FM | **0.689** | 0.549 | 0.640 | 0.642 | **0.645** | 0.407 | 0.593 | 0.382 | 0.167 |
| IV | **0.715** | 0.618 | **0.730** | 0.666 | 0.648 | 0.533 | 0.663 | 0.386 | 0.131 |
| IPR | **0.682** | 0.555 | **0.666** | 0.603 | 0.631 | 0.471 | 0.596 | 0.397 | 0.203 |
| LR | **0.638** | 0.352 | 0.554 | **0.616** | 0.597 | 0.531 | 0.469 | 0.253 | 0.038 |
| MB | **0.722** | 0.608 | **0.667** | 0.634 | 0.659 | 0.410 | 0.597 | 0.335 | 0.203 |
| OCC | **0.648** | 0.528 | **0.651** | 0.625 | 0.636 | 0.494 | 0.596 | 0.355 | 0.110 |
| OPR | **0.672** | 0.601 | 0.639 | **0.666** | 0.650 | 0.499 | 0.584 | 0.372 | 0.190 |
| OV | **0.801** | 0.646 | **0.748** | 0.685 | 0.674 | 0.368 | 0.550 | 0.400 | 0.204 |
| SV | **0.699** | 0.570 | **0.666** | 0.622 | 0.629 | 0.500 | 0.594 | 0.388 | 0.144 |



(a) (b)

Fig. 9. Precision and success plots of all the compared hyperspectral trackers on the IMEC25 dataset. (a) Precision plot. (b) Success plot.

TABLE V
IMPACT OF DIFFERENT VIEWS ON TRACKING PERFORMANCE

| False-color | Material | Hyperspectral | AUC | DP |
|---|---|---|---|---|
| ✓ | ✗ | ✗ | 0.634 | 0.865 |
| ✗ | ✓ | ✗ | 0.533 | 0.769 |
| ✗ | ✗ | ✓ | 0.642 | 0.876 |
| ✓ | ✓ | ✗ | 0.645 | 0.884 |
| ✗ | ✓ | ✓ | 0.652 | 0.891 |
| ✓ | ✓ | ✓ | **0.661** | **0.902** |

TABLE VI
EFFECTIVENESS OF THE PROPOSED KEY COMPONENTS

| Components | | | | Indexes | |
|---|---|---|---|---|---|
| CBA | CMA | IVIM | MGCVF | AUC | DP |
| ✓ | ✓ | ✓ | ✓ | **0.691** | **0.932** |
| ✓ | ✗ | ✓ | ✓ | 0.681 | 0.915 |
| ✗ | ✓ | ✓ | ✓ | 0.676 | 0.913 |
| ✗ | ✗ | ✓ | ✓ | 0.668 | 0.902 |
| ✓ | ✓ | ✗ | ✓ | 0.676 | 0.923 |
| ✓ | ✓ | ✓ | ✗ | 0.670 | 0.896 |
| ✓ | ✓ | ✗ | ✗ | 0.661 | 0.902 |

### *C. Ablation Study and Parameter Analysis*

This section studies the impact of multiple view exploitation, all the introduced components, and the number of abundance maps on the tracking performance. Moreover, we study the impact of different views as guidance to combine other views on tracking performance. We also conducted experiments to study the impact of hyperparameters on the tracking performance. All the experiments were performed on the HOTC dataset.

*1) Effectiveness of Using Multiview Representation:* In this experiment, we examine the effectiveness of multiview information for tracking. For multiview fusion, the feature maps from various views are directly summed to eliminate the impact of the fusion strategy. As the network parameters are significantly fewer than the original network, a reduced learning rate is utilized. In particular, the learning rate is linearly increased from $1 \times 10^{-6}$ to $5 \times 10^{-5}$ during the first epoch. Following that, a cosine annealing learning rate schedule is adopted to decrease the learning rate from $5 \times 10^{-5}$ to $1 \times 10^{-6}$ over the remaining 19 epochs. The other settings were the same with MMF-Net. The tracking outcomes are given in Table V. The outcomes suggest that multiview outperforms a single view since it can provide a more comprehensive object description. Additionally, combining the material view with either a false-color or hyperspectral view improves the tracking performance, demonstrating that material information adds extra discrimination to handle challenging scenarios. It is worth noting that the combination of three views produces the highest tracking accuracy.

*2) Effectiveness of the Proposed Key Components:* In these experiments, we analyze the effectiveness of proposed key modules for tracking performance. The original MMF-Net tracker was modified to show effectiveness by dropping each module. Table VI shows that removing the CBA and CMA blocks results in reduced tracking performance due to the lack of cross-material and cross-band interaction. Similarly, the removal of IVIM reduces the AUC to 0.676, as it fails to establish global visual contextual connections within a single view. Additionally, dropping the MGCVF can negatively impact the tracking performance by losing the benefits of cross-view complementarity. Furthermore, in Fig. 11, we include the classification response maps of various models generated by dropping different blocks. As depicted, removing different blocks leads to unreliable sub-peaks in all cases. Notably, our method's response map suppresses the distracting subpeak and produces a sharper target peak by utilizing all modules. Overall, this experiment demonstrates the effectiveness of the introduced components.

*3) Impact of Different Views as Guidance:* To evaluate the effectiveness of the material view, we conducted ablation experiments where we replaced the material view with the hyperspectral view and false-color view as guidance for fusion. The tracking results are presented in Fig. 12. It can be observed that the tracking performance degrades when using either hyperspectral or false-color information as guidance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

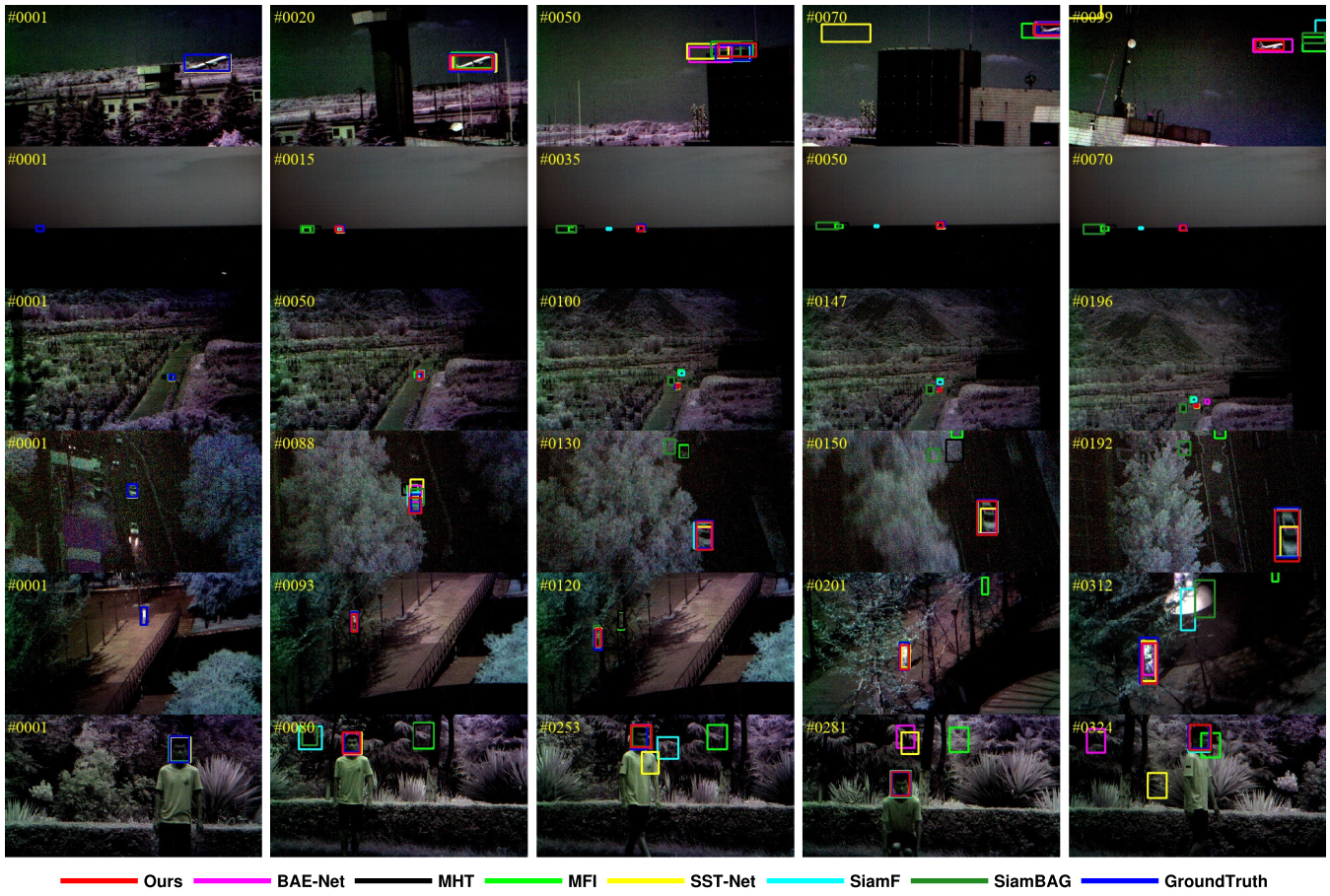



Fig. 10. Qualitative comparison of the MMF-Net tracker on six video sequences (*airplane4, boat, doublecar, doublecar8, human3, and man3*) on the IMEC25 dataset. The proposed approach performs favorably against these trackers (for better illustration, we show the bounding boxes on the false-color image converted from HSIs).

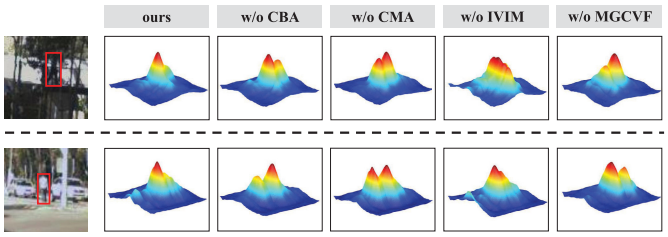
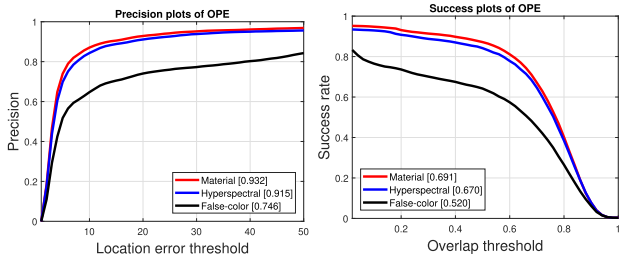
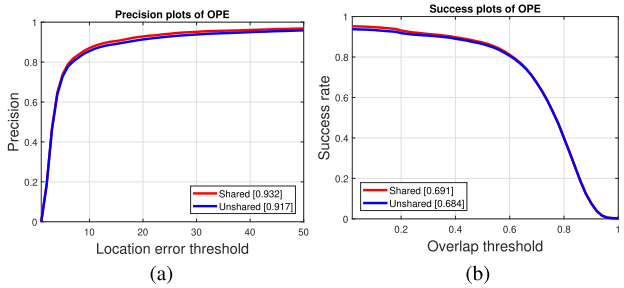



Fig. 11. Visualization of classification response maps when dropping different blocks.




(a) (b)

Fig. 12. Impact of different views as guidance on the tracking performance. (a) Precision plot. (b) Success plot.




(a) (b)

Fig. 13. Impact of shared and unshared backbone networks on the tracking performance. (a) Precision plot. (b) Success plot.

Moreover, using the false-color view as guidance results in the most inferior performance because of the absence of the material identification ability. Overall, the results from these ablation experiments evidently validate the effectiveness of incorporating the material view as guidance for cross-view fusion in hyperspectral object tracking.

*4) Impact of the Backbone Network Setting:* Here, we modified the shared backbone network across views to an unshared one. This adjustment leads to an increased number of parameters to be learned. Consequently, an unshared backbone across different views results in lower tracking accuracy, as depicted in Fig. 13.

*5) Impact of Different Hyperparameters:* We investigated the impact of various hyperparameters on tracking performance, including the number of heads in multihead attention, scale factor $\tau$ in CBA and CMA blocks, hyperparameters in loss function, and the number of materials. First, we explored the effect of the number of heads in multihead attention, ranging from 1 to 16. As shown in Fig. 14, we determine

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

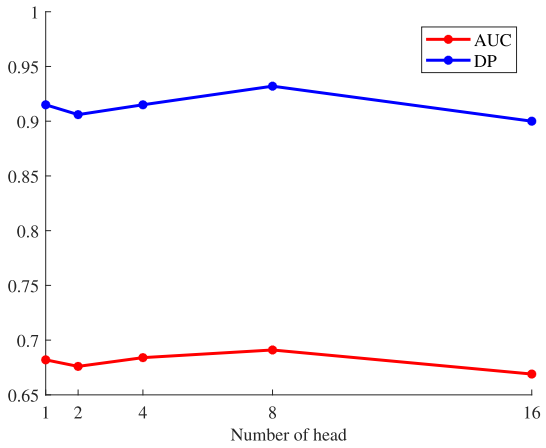



Fig. 14. Impact of head number in the multihead attention on the tracking performance.

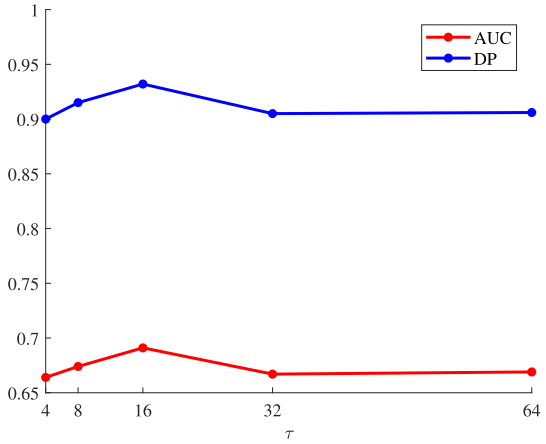



Fig. 15. Impact of $\tau$ on the tracking performance.

TABLE VII
IMPACT OF DIFFERENT VALUES OF $\lambda_3$ ON TRACKING PERFORMANCE

| $\lambda_3$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| AUC | 0.685 | 0.686 | **0.691** | 0.665 | 0.658 |
| DP | 0.920 | 0.923 | **0.932** | 0.890 | 0.891 |

that the optimal tracking performance was achieved with eight heads. Next, we examined the scale factor $\tau$ in CBA and CMA blocks. By varying $\tau$ from 4 to 64, we discover that the best tracking performance was attained when $\tau = 16$, as shown in Fig. 15. Based on this finding, we set $\tau$ to 16. As for the hyperparameters in the loss function, we follow the default settings in SiamFC++ [65] and STMTrack [55] to determine $\lambda_1$ and $\lambda_2$. Moreover, we explored the impact of different values for $\lambda_3$ on tracking performance. The results presented in Table VII indicate that the best tracking performance is achieved when $\lambda_3 = 3$.

In addition, we examined how the number of materials affects the tracking performance. Specifically, we varied the value from 2 to 10, with an interval of 2. The results presented in Fig. 16 indicate that setting the number to six yields the best tracking performance. Hence, we adopt six as the number of abundance maps in our experiments.

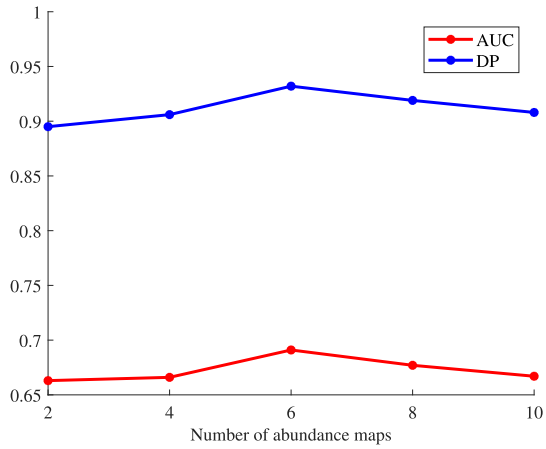



Fig. 16. Impact of different numbers of abundance maps in material view.

## V. CONCLUSION

In this article, we have introduced an MMF-Net for hyperspectral object tracking. Thanks to learning the interaction across different materials, our CMAs are beneficial to improve the representation ability of an individual view. Moreover, our material-guided multiview fusion module is proven to effectively mine multiview complement to establish a joint robust representation, which enables accurate object location. Extensive experiments on HOTC, HOT2022, and IMEC25 datasets show that our MMF-Net achieves excellent tracking results. Notably, our MMF-Net does not achieve optimal tracking on some challenging attributes such as OCC and IV. In future work, we plan to integrate temporal interaction information between adjacent frames to overcome this issue.

## REFERENCES

[1] Y. Cui, B. Hou, Q. Wu, B. Ren, S. Wang, and L. Jiao, "Remote sensing object tracking with deep reinforcement learning under occlusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605213.

[2] W. Song et al., "A joint Siamese attention-aware network for vehicle object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625617.

[3] M. Danelljan, G. Häger, F. S. Khan, and M. Felsberg, "Discriminative scale space tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 8, pp. 1561–1575, Aug. 2017.

[4] A. Lukezic, T. Vojir, L. Cehovin Zajc, J. Matas, and M. Kristan, "Discriminative correlation filter with channel and spatial reliability," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2017, pp. 6309–6318.

[5] Z. Chen, B. Zhong, G. Li, S. Zhang, and R. Ji, "Siamese box adaptive network for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6668–6677.

[6] F. Tang and Q. Ling, "Ranking-based Siamese visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8741–8750.

[7] X. Wu, W. Li, D. Hong, R. Tao, and Q. Du, "Deep learning for unmanned aerial vehicle-based object detection and tracking: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 91–124, Mar. 2022.

[8] M. Zhao, S. Li, S. Xuan, L. Kou, S. Gong, and Z. Zhou, "SatSOT: A benchmark dataset for satellite video single object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617611.

[9] J. Shao, B. Du, C. Wu, M. Gong, and T. Liu, "HRSiam: High-resolution Siamese network, towards space-borne satellite video tracking," *IEEE Trans. Image Process.*, vol. 30, pp. 3056–3068, 2021.

[10] S. Xuan, S. Li, M. Han, X. Wan, and G. S. Xia, "Object tracking in satellite videos by improved correlation filters with motion estimations," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1074–1086, Feb. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

[11] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[12] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2017, pp. 233–242.

[13] Z. Liu, Y. Zhong, X. Wang, M. Shu, and L. Zhang, "Unsupervised deep hyperspectral video target tracking and high spectral–spatial–temporal resolution ($H^3$) benchmark dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513814.

[14] Z. Hou, W. Li, J. Zhou, and R. Tao, "Spatial–spectral weighted and regularized tensor sparse correlation filter for object tracking in hyperspectral videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541012.

[15] Y. Zhang, X. Li, B. Wei, L. Li, and S. Yue, "A fast hyperspectral tracking method via channel selection," *Remote Sens.*, vol. 15, no. 6, p. 1557, Mar. 2023.

[16] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "BAE-Net: A band attention aware ensemble network for hyperspectral object tracking," in *Proc. IEEE ICIP*, Oct. 2020, pp. 2106–2110.

[17] Z. Li, X. Ye, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Spectral–spatial–temporal attention network for hyperspectral tracking," in *Proc. IEEE Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[18] Z. Liu, X. Wang, Y. Zhong, M. Shu, and C. Sun, "SiamHYPER: Learning a hyperspectral object tracker from an RGB-based tracker," *IEEE Trans. Image Process.*, vol. 31, pp. 7116–7129, 2022.

[19] J. Lei, P. Liu, W. Xie, L. Gao, Y. Li, and Q. Du, "Spatial–spectral cross-correlation embedded dual-transfer network for object tracking using hyperspectral videos," *Remote Sens.*, vol. 14, no. 15, p. 3512, Jul. 2022.

[20] Z. Liu et al., "An anchor-free Siamese target tracking network for hyperspectral video," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[21] Y. Wang, Y. Liu, M. Ma, and S. Mei, "A spectral–spatial transformer fusion method for hyperspectral video tracking," *Remote Sens.*, vol. 15, no. 7, p. 1735, Mar. 2023.

[22] W. Li, Z. Hou, J. Zhou, and R. Tao, "SiamBAG: Band attention grouping-based Siamese object tracking network for hyperspectral videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514712.

[23] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[24] Z. Zhang, S. Liao, H. Fang, H. Zhang, and S. Wang, "Sparse hyperspectral unmixing using spectral library adaptive adjustment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4873–4887, Dec. 2019.

[25] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 818–833.

[26] S. L. Al-Khafaji, J. Zhou, X. Bai, Y. Qian, and A. W. Liew, "Spectral–spatial boundary detection in hyperspectral images," *IEEE Trans. Image Process.*, vol. 31, pp. 499–512, 2022.

[27] J. Liang, J. Zhou, L. Tong, X. Bai, and B. Wang, "Material based salient object detection from hyperspectral images," *Pattern Recognit.*, vol. 76, pp. 476–490, Apr. 2018.

[28] J. Qi, Z. Gong, W. Xue, X. Liu, A. Yao, and P. Zhong, "An unmixing-based network for underwater target detection from hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5470–5487, 2021.

[29] Z. Li, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Material-guided Siamese fusion network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 2809–2813.

[30] F. Xiong, J. Zhou, W. Charle, Y. Zhong, P. Ghamisi, and J. Chanussot. (2022). *The Hyperspectral Object Tracking Challenge (HOT2022)*. [Online]. Available: https://www.hsitracking.com

[31] L. Chen, Y. Zhao, J. C.-W. Chan, and S. G. Kong, "Histograms of oriented mosaic gradients for snapshot spectral image description," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 79–93, Jan. 2022.

[32] K. Qian, J. Zhou, F. Xiong, H. Zhou, and J. Du, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *Proc. Int. Conf. Smart Multimedia (ICSM)*. Cham, Switzerland: Springer, 2018, pp. 308–319.

[33] Z. Li, F. Xiong, J. Zhou, J. Lu, and Y. Qian, "Learning a deep ensemble network with band importance for hyperspectral object tracking," *IEEE Trans. Image Process.*, vol. 32, pp. 2901–2914, 2023.

[34] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Tracking in aerial hyperspectral videos using deep kernelized correlation filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 449–461, Jan. 2019.

[35] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015, pp. 1–14.

[36] L. Gao et al., "CBFF-Net: A new framework for efficient and accurate hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506114.

[37] Y. Li, M. Yang, and Z. Zhang, "A survey of multi-view representation learning," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 10, pp. 1863–1883, Oct. 2019.

[38] M. Xu, Q. Zhao, and S. Jia, "Multiview spatial–spectral active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512415.

[39] X. Song, J. Chen, Z. Wu, and Y.-G. Jiang, "Spatial–temporal graphs for cross-modal Text2Video retrieval," *IEEE Trans. Multimedia*, vol. 24, pp. 2914–2923, 2022.

[40] Y. Zhao, W. Wang, H. Zhang, and B. Hu, "Learning homogeneous and heterogeneous co-occurrences for unsupervised cross-modal retrieval," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2021, pp. 1–6.

[41] Q. Gao, H. Lian, Q. Wang, and G. Sun, "Cross-modal subspace clustering via deep canonical correlation analysis," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 3938–3945.

[42] Z. Sun, P. Sarma, W. Sethares, and Y. Liang, "Learning relationships between text, audio, and video via deep canonical correlation for multimodal language analysis," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 5, pp. 8992–8999.

[43] G. Andrew, R. Arora, J. Bilmes, and K. Livescu, "Deep canonical correlation analysis," in *Proc. ICML*, 2013, pp. 1247–1255.

[44] C. Zhang, Y. Liu, and H. Fu, "AE2-Nets: Autoencoder in autoencoder networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2572–2580.

[45] J. Li, Y. Ma, R. Song, B. Xi, D. Hong, and Q. Du, "A triplet semisupervised deep network for fusion classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540513.

[46] H. Fazlali, Y. Xu, Y. Ren, and B. Liu, "A versatile multi-view framework for LiDAR-based 3D object detection with guidance from panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17171–17180.

[47] Y. Liu et al., "Where to focus: Investigating hierarchical attention relationship for fine-grained visual classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2022, pp. 57–73.

[48] Y. Deng, H. Chen, and Y. Li, "MVF-Net: A multi-view fusion network for event-based object classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 12, pp. 8275–8284, Dec. 2022.

[49] R. Fan, F. Li, W. Han, J. Yan, J. Li, and L. Wang, "Fine-scale urban informal settlements mapping by fusing remote sensing images and building data via a transformer-based multimodal fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630316.

[50] B. Yan, H. Peng, J. Fu, D. Wang, and H. Lu, "Learning spatio-temporal transformer for visual tracking," in *Proc. Int. Conf. Comput. Vision. (ICCV)*, 2021, pp. 10448–10457.

[51] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, 2017, pp. 1–11.

[52] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[53] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604513.

[54] Y. Bazi, M. M. A. Rahhal, M. L. Mekhalfi, M. A. A. Zuair, and F. Melgani, "Bi-modal transformer-based approach for visual question answering in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708011.

[55] Z. Fu, Q. Liu, Z. Fu, and Y. Wang, "STMTrack: Template-free visual tracking with space-time memory networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13774–13783.

[56] C. Szegedy et al, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

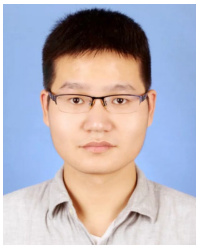

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.

[57] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.
[58] R. Heylen, D. Burazerovic, and P. Scheunders, "Fully constrained least squares spectral unmixing by simplex projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4112–4122, Nov. 2011.
[59] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 213–229.
[60] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2980–2988.
[61] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "Unitbox: An advanced object detection network," in *Proc. ACM Int. Conf. Multimedia*, 2016, pp. 516–520.
[62] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.
[63] K. Zhang, Q. Liu, Y. Wu, and M.-H. Yang, "Robust visual tracking via convolutional networks without training," *IEEE Trans. Image Process.*, vol. 25, no. 4, pp. 1779–1792, Apr. 2016.
[64] Z. Zhang, K. Qian, J. Du, and H. Zhou, "Multi-features integration based hyperspectral videos tracker," in *Proc. WHISPERS*, Mar. 2021, pp. 1–5.
[65] Y. Xu, Z. Wang, Z. Li, Y. Yuan, and G. Yu, "SiamFC++: Towards robust and accurate visual tracking with target estimation guidelines," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12549–12556.



**Zhuanfeng Li** received the B.E. degree in computer science and technology from Wuxi University, Wuxi, China, in 2018. He is currently pursuing the Ph.D. degree in computer science and technology with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

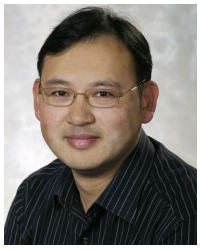
His research interests include hyperspectral video tracking and machine learning.



**Fengchao Xiong** (Member, IEEE) received the B.E. degree in software engineering from Shandong University, Jinan, China, in 2014, and the Ph.D. degree from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2019.

He visited Wuhan University, Wuhan, China; Griffith University, Nathan, QLD, Australia; and the University of Macau, Taipa, Macau, China, from 2011 to 2012, 2017 to 2018, and 2021 to 2023, respectively. He is currently an Associate Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. His research interests include hyperspectral image processing, machine learning, and pattern recognition.

Dr. Xiong serves as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



**Jun Zhou** (Senior Member, IEEE) received the B.S. degree in computer science and the B.E. degree in international business from Nanjing University of Science and Technology, Nanjing, China, in 1996 and 1998, respectively, the M.S. degree in computer science from Concordia University, Montreal, QC, Canada, in 2002, and the Ph.D. degree from the University of Alberta, Edmonton, AB, Canada, in 2006.

In June 2012, he joined the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, where he is currently a Professor. Before this appointment, he was a Research Fellow with the Research School of Computer Science, Australian National University, Canberra, ACT, Australia, and a Researcher with the Canberra Research Laboratory, NICTA, Canberra. His research interests include pattern recognition, computer vision, and spectral imaging with their applications in remote sensing and environmental informatics.

Dr. Zhou is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition* journal.



**Jianfeng Lu** (Member, IEEE) received the M.S. and Ph.D. degrees in pattern recognition and intelligent systems from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China, in 1994 and 2000, respectively.

He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. His research interests include image analysis, machine learning, data mining, and intelligent robots.



**Zhuang Zhao** received the Ph.D. degree in optical engineering from Nanjing University of Science and Technology, Nanjing, China, in 2019.

He is currently an Associate Professor with the School of Electronic Engineering and Optoelectronic Technology, Nanjing University of Science and Technology. His research interests include spectral image reconstruction and snapshot spectrum detection.



**Yuntao Qian** (Senior Member, IEEE) received the B.E. and M.E. degrees in automatic control from Xi'an Jiaotong University, Xi'an, China, in 1989 and 1992, respectively, and the Ph.D. degree in signal processing from Xidian University, Xi'an, in 1996.

From 1996 to 1998, he was a Post-Doctoral Fellow with Northwestern Polytechnical University, Xi'an. Since 1998, he has been with the College of Computer Science, Zhejiang University, Hangzhou, China, where he became a Professor in 2002. During 1999–2001, 2006, 2010, 2013, 2015–2016, and 2018, he was a Visiting Professor at Concordia University, Montreal, QC, Canada; Hong Kong Baptist University, Hong Kong; Carnegie Mellon University, Pittsburgh, PA, USA; the Canberra Research Laboratory, NICTA, Canberra, ACT, Canada; Macau University, Macau, China; and Griffith University, Nathan, QLD, Australia. His research interests include machine learning, signal and image processing, pattern recognition, and hyperspectral imaging.

Dr. Qian is currently an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 17:31:43 UTC from IEEE Xplore. Restrictions apply.